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Portable light with multiple operating modes

Abstract

A portable light is configured to be operated in a range of selectable operational modes to provide flexible illumination solutions in both stationary and portable situations. The light includes an elongated central body and a power source positioned within the elongated central body. The light includes an operating mode selector assembly to control which lighting elements are illuminated during operation of the portable light and a luminosity selector assembly to selectively control the lumen output (brightness) of the lighting elements.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/316,411, which is a continuation of U.S. Pat. No. 11,678,414, which is a continuation of U.S. Pat. No. 10,914,433, which is a continuation of U.S. Pat. No. 10,598,319, which is a continuation of U.S. Pat. No. 10,371,326, which is a continuation-in-part of U.S. Pat. No. D923,836, and which claims the benefit of Provisional Patent Application No. 62/549,247, all of which are incorporated in their entirety herein by reference and made a part hereof.

TECHNICAL FIELD

(1) This disclosure relates to a portable light with a central battery that has multiple operating modes and multiple mounting configurations. In particular, the portable light includes one or more

light-emitting elements selectively operable among a plurality of modes.

BACKGROUND OF THE INVENTION

(2) Existing lighting products enables a wide range of indoor, outdoor and nighttime activities. Electronic lighting is typically provided from fixed installations (e.g., a roof, a wall, or ceiling), where a light source receives electrical power from a fixed and wired power source. Such lighting is useful in illuminating a particular area, but lacks the flexibility of a portable lighting source. Internally-powered portable lighting sources have been developed to provide illumination in varied locations and situations. However, such internally-powered portable lighting sources are not optimized for size, durability, form-factor, illumination flexibility and mounting flexibility parameters. Further, portable lighting sources are also not optimized to provide illumination both in fixed locations and in varied locations.

(3) Accordingly, there is an unmet need for a portable light optimized for durability and to provide a variety of illumination modes and mounting configurations. A full discussion of the features and advantages of the present invention is deferred to the following detailed description, which proceeds with reference to the accompanying drawings.

SUMMARY OF THE INVENTION

(4) In some implementations, the present disclosure provides a portable light that features a plurality of lighting elements, an internal power source, a mode selector for selecting among various operating modes, a luminosity selector for selecting among various lighting element brightness levels, a magnet, a securement member selectively positionable among multiple configurations, and a retractable stabilization assembly.

(5) Other features and advantages of the invention will be apparent from the following specification taken in conjunction with the following drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements.

(2) FIG. 1 is a top perspective view of a first embodiment of a portable light, showing a stabilization assembly in a retracted position and a securement member in a deployed position.

(3) FIG. 2 is a top perspective view of the portable light source of FIG. 1, wherein the portable light is resting on a support surface with both the stabilization assembly and the securement member in a deployed position.

(4) FIG. 3 is a bottom perspective view of the portable light of FIG. 1, wherein both the stabilization assembly and the securement member are in the retracted position.

(5) FIG. 4 is a front view of the portable light of FIG. 1, wherein both the stabilization assembly and the securement member are in the retracted position.

(6) FIG. 5 is a rear view of the portable light of FIG. 1, wherein both the stabilization assembly and the securement member are in the retracted position.

(7) FIG. 6 is a top view of the portable light of FIG. 1, wherein both the stabilization assembly and the securement member are in the retracted position.

(8) FIG. 7 is a bottom view of the portable light of FIG. 1, wherein both the stabilization assembly and the securement member are in the retracted position.

(9) FIG. 8 is a perspective view of the portable light of FIG. 1, wherein the portable light is in a partially disassembled state.

(10) FIG. 9 is a cross-sectional view of the portable light of FIG. 1 taken along the 9-9 line shown in FIG. 4.

- (11) FIG. 9A is an enlarged cross-sectional view of the first end assembly of the portable light of FIG. 1 taken along the 9-9 line shown in FIG. 4.
- (12) FIG. 10 is a cross-sectional view with perspective treatment of the portable light shown in FIG. 1 taken along line 10-10 of FIG. 4.
- (13) FIG. 11 is a cross-sectional view with perspective treatment of the portable light shown in FIG. 1 taken along line 11-11 of FIG. 4.
- (14) FIG. 12 is a block diagram of the circuitry in the portable light of FIG. 1.
- (15) FIG. 13 is an exploded view of the portable light shown in FIG. 1.
- (16) FIG. 14 is a top perspective view of a second embodiment of a portable light, showing a securement assembly in a retracted position.
- (17) FIG. 15 is a bottom perspective view of the portable light of FIG. 14, wherein the securement assembly is in the retracted position.
- (18) FIG. 16 is a front view of the portable light of FIG. 14, wherein the securement assembly is in the retracted position.
- (19) FIG. 17 is a rear view of the portable light of FIG. 14, wherein the securement assembly is in the retracted position.
- (20) FIG. 18 is a bottom view of the portable light of FIG. 14, wherein the securement assembly is in the retracted position.
- (21) FIG. 19 is a top view of the portable light of FIG. 14, wherein the securement assembly is in the retracted position.
- (22) FIG. 20 is a bottom perspective view of the portable light of FIG. 14, showing the securement assembly is in a deployed position.
- (23) FIG. 21 is a top view of the portable light of FIG. 14, wherein the securement assembly is in the retracted position, a mode selector in a first position, and a single lighting element is in an illuminating state.
- (24) FIG. 22 is a top view of the portable light source of FIG. 14, wherein the securement assembly is in the retracted position, a mode selector in a fourth position, and all four lighting elements are in an illuminating state.
- (25) FIG. 23 is a top perspective view of a third embodiment of a portable light, showing a stabilization assembly in a retracted position and a securement member in a retracted position.
- (26) FIG. 24 is a bottom perspective view of the portable light shown in FIG. 23, showing the stabilization assembly in the retracted position and a securement member in a deployed position.
- (27) FIG. 25 is a front view of the portable light source shown in FIG. 23, wherein the securement member is in a deployed position.
- (28) FIG. 26 is a perspective view of the power source for use with the portable light shown in embodiments 1-3.
- (29) FIG. 27 is a block diagram of the circuitry in the power source and the portable light of embodiments 1-3 in a first direction.
- (30) FIG. 28 is a block diagram of the circuitry in the power source and the portable light of embodiments 1-3 in a second direction.
- (31) FIG. 29 is a rear view of the portable light of FIG. 1, wherein the second end assembly has been removed from the portable light 10 and replaced with a flashlight head 300.
- (32) While the invention will be described in connection with the preferred embodiments shown herein, it will be understood that it is not intended to limit the invention to those embodiments. On the contrary, it is intended to cover all alternatives, modifications, and equivalents, as may be included within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

(33) While this disclosure includes a number of details and implementations in many different forms, there is shown in the drawings and will herein be described in detail particular implementations with the understanding that the present disclosure is to be considered as an

exemplification of the principles of the disclosed methods and systems, and is not intended to limit the broad aspects of the disclosed concepts to the implementations illustrated.

(34) As shown in the Figures and explained below, a portable light or lantern **10** features a multitude of brightness or luminosity settings and a distinct operating modes, both independently selected by a user. The portable light **10** can be carried or mounted using a plurality of mounting configurations (e.g., on a flat support surface **44**) in order to provide flexible illumination solutions to the user. The multiple operating modes and the multiple mounting configurations provide the portable light **10** with increased flexibility and greater utility, as compared to conventional lanterns or flashlights.

(35) With reference to the Figures, the portable light **10** includes: (i) a luminosity selector assembly **28**, (ii) an operating mode selector assembly **32**, (iii) a plurality of lighting elements **100**, (iv) a power source **134** disposed between the lighting elements **100**, and (v) a retractable stabilization assembly **192**. As discussed in greater detail below, the luminosity selector assembly **28** and the mode selector assembly **32** allow the user to independently change the operation mode and the brightness or lumen output level of the portable light **10**. These assemblies **28**, **32** operate independent of each other, where the user: (i) selects at least one lighting element **100** to be illuminated using the mode selector assembly **32** and (ii) moves the luminosity selector assembly **28** from an “Off” position to an “On” position. Specifically, the luminosity selector assembly **28** allows the user to set the brightness or lumen output level of the lighting element(s) **100**, such that the luminosity selector assembly **28** functions as a dimmer for the portable light **10**. This dimming function allows the user to select a brightness level or lumen output from the portable light **10**, which is between a first predetermined minimum level and a second predetermined maximum level. The mode selector assembly **32** allows the user to select the operating mode of the portable light **10**. In particular, the mode selector assembly **32** allows the user to select the number of lighting elements **100** to be illuminated. For example, the user may select one of the lighting elements **100** or a plurality of lighting elements **100** for illumination.

(36) As generally shown in FIGS. **8-11**, the portable light **10** also has a power source **134** that is disposed within the intermediate section **22** and thus radially inward of the lighting elements **100**. The power source **134** may be a combination of removable non-rechargeable batteries **278** disposed within a battery cartridge **250**. As discussed in greater detail below, the configuration of the power source **134** allows it to supply power to the portable light **10** regardless of which direction the power source **134** is inserted within the portable light **10**.

(37) The portable light **10** further has multiple mounting configurations for use in different environments and under different conditions. For example, the user can move the securement member **30** to a deployed position **214**, as best shown in FIGS. **1** and **2**, where upon the user can then hang the portable light **10** via the deployed securement member **30** to an object or fixture (e.g., a hook) disposed above the portable light **10**. Second, the user may releasably attach the magnet **190** residing within the lower end **18** to a magnetic surface or object. Thus, the portable light **10** will be magnetically mounted, to a selected magnetic surface or object at any orientation (e.g., to a tool box or to a metal surface of a vehicle). Third, the user may place the bottom surface **199** of the portable light **10** on a surface support surface **44** and deploy a retractable stabilization assembly **192** (see FIG. **2**). Fourth, the user may manually carry the portable light **10** for portable illumination use as a flashlight or lantern between various locations.

(38) In the embodiment of FIGS. **1-12**, the portable light **10** has an elongated configuration with a first end assembly **24** that is operably connected to the first end **14** of the portable light **10**. The first end assembly **24** allows the user to change the operating mode and the brightness of the portable light **10** by changing the positions of the luminosity selector assembly **28** and mode selector assembly **32**. Referring to FIGS. **1-5**, the first end assembly **24** includes: (i) a top or upper housing **26**, (ii) the luminosity selector assembly **28**, (iii) the securement member **30**, (iv) the mode selector assembly **32**, and (v) connector rings **34**, **36**. The upper housing **26** has an end wall **38** that is

inwardly recessed from an outer rim **40** to form a recess **42**. An extent of the luminosity selector assembly **28** is positioned within the recess **42**. Referring to FIG. **9**, the luminosity selector assembly **28** preferably does not extend beyond the outer rim **40**, which helps protect the luminosity selector assembly **28** from making contact with a support surface **44**, if the portable light **10** is dropped or knocked over by the user. In addition, this configuration allows the outer rim **40** to make contact with a support surface **44** without interruption by the luminosity selector assembly **28**, which in turn allows the portable light **10** to be stable, if the outer rim **40** is placed in contact with the support surface **44**.

(39) As mentioned above, the luminosity selector assembly **28** functions as a dimmer for the portable light **10**. This dimming function allows the user to select a brightness level or lumen output from the lighting elements **100**, which is between a first predetermined minimum level (e.g., 20 lumen) and a second predetermined maximum level (3,000 lumen). In the embodiment of the Figures, the luminosity selector assembly **28** is a rotary dial **46** that includes: (i) a cap **48**, (ii) a selector **50**, (iii) an o-ring **52**, and (iv) a luminosity printed circuit board (“PCB”) **54**. Referring to FIGS. **9** and **13**, the cap **48** includes a flange **56** and a projection **58** that extends radially from the flange **56**. The cap **48** is configured such that the projection **58** extends through a hole **60** formed in the end wall **38**, while the flange **56** resides below or within the end wall **38**. This configuration helps to ensure that when an upwardly directed force that is applied to the cap **48**, the cap **48** cannot be removed from the portable light **10**. The o-ring **52** is positioned between the flange **56** and the end wall **38**, which helps to ensure the smooth operation of the rotary dial **46** along with protecting the internal electronics from exterior moisture.

(40) The cap **48** is operably coupled to the selector **50**, which in turn is operably coupled to the luminosity PCB **54**. Thus, when the cap **48** is rotated by a user, the selector **50** is also rotated. This angular movement of the cap **48** and the selector **50** changes the state of the luminosity PCB **54**, which in turn changes the current that may be applied to the lighting elements **100** during operation of the light **10**. This change in current alters the lumen output of the lighting elements **100**. For example, the rotary dial **46** may be rotated counter-clockwise until it stops. In this “OFF” position, the luminosity PCB **54** applies zero electrical current to the lighting elements **100**. As the rotary dial **46** is rotated in a clockwise direction from this “OFF” position, the luminosity PCB **54** allows for the application of a greater amount of electrical current to the lighting elements **100**. This in turn increases the possible lumen output of the lighting elements **100** (assuming the mode selector assembly **32** is not in the “Off” position).

(41) In some embodiments (see FIGS. **14** and **19**) the rotary dial **46** may include dial grooves **61** or another type of textured surface to enhance a user’s grip when rotating the rotary dial **46**. In an alternative embodiment, the luminosity selector assembly **28** may have multiple predetermined lumen output levels. For example, the luminosity selector assembly **28** may be set to a first or “OFF” position, a second or “Low” position, a third or “Medium” position, and a fourth or “High” position. It should be understood that the luminosity selector assembly **28** may have other mechanical or electrical configurations, which may further its ability to function as a dimmer for the portable light **10**.

(42) It is to be understood that the luminosity selector assembly **28** can be replaced with an assembly that is configured to utilize one or more buttons, switches, sliders, local sensors (e.g., motion, light, sound, heat, motion), or other types of electrical selection devices. In one embodiment, the rotary dial **46** and its associated circuitry may be replaced by a button and the necessary circuitry to allow a user to illuminate or turn “On” the portable light **10** upon depressing the button and to un-illuminate or turn “Off” the portable light **10** upon depressing the button another time after the portable light **10** is in an illuminated state. In another alternative embodiment, the rotary dial **46** and its associated circuitry may be replaced by a button and the necessary circuitry to allow a user to either: (i) press the button a first time to illuminate or turn “On” the portable light **10**, (ii) press and hold the button to illuminate and adjust the brightness of

the portable light **10**, or (iii) press the button another time after the portable light **10** is illuminated to un-illuminate or turn “Off” the light **10**. Additionally, the luminosity selector assembly **28** may be remotely located from the portable light **10**. For example, the luminosity selector assembly **28** may be located on a remote wireless device, such as the remote devices described in U.S. patent application Ser. No. 15/812,852, filed on May 17, 2018, and which is fully incorporated herein by reference. Another example of a remote wireless device includes a cellphone, laptop, RF remote control, or other devices that are connected to the portable light **10** via the internet (e.g., wireless camera, motion sensor, light sensor, timer, etc.).

(43) As best shown in FIGS. **1** and **6**, a luminosity indicator **62** may be disposed on the end wall **38** or the luminosity selector assembly **28**. The luminosity indicator **62** contains visual information including shapes, letters or numbers for apprising a user of information regarding an operation of the luminosity selector assembly **28**. In some implementations, as shown exemplarily in FIG. **6**, the luminosity indicator **62** includes the word “OFF” **64** indicating zero luminosity, along with a visual image indicating progressively increased luminosity **66**. In other embodiments, the luminosity indicator **62** may have multiple words (e.g., “OFF”, “Low”, “Medium”, and “High”), may be made from a reflective material, can be a light emitting diode, or a TFT display.

(44) The upper housing **26** also contains components that help position and secure the electrical components of the portable light **10** within the first end assembly **24**. As best shown in FIGS. **1**, **9** and **13**, the upper housing **26** also includes: (i) a notch **68**, (ii) mode selector aperture **70**, (iii) a first set of internal threads **72**, (iv) a second set of internal threads **74**, (v) a first set of external threads **76**, and (vi) a upper flange **78**. The notch **68** receives an extent of an O-ring **80** that is positioned between the upper housing **26** and a mode selector ring **82** of the mode selector assembly **32**. This configuration helps maintain spacing between the mode selector ring **82** and the outer surface of the upper housing **26** to ensure the smooth operation of the mode selector ring **82**. The first set of internal threads **72** has a smaller internal diameter than the second set of threads **74**, wherein the first set of internal threads **72** are configured to receive an extent of a luminosity PCB **54** and a mode selector PCB **84**. This configuration allows for quick assembly of the portable light **10** and the replacement of the luminosity PCB **54** or the mode selector PCB **84**, if either one of these PCBs are damaged. The first set of external threads **76** are configured to receive the internal threads **75** of the connector rings **34**, **36**. As will be described in greater detail below, the mode selector PCB **84** helps control which of the lighting elements **100** is illuminated.

(45) The first end assembly **24** also includes a first spacer **86** and a second spacer **88**, which are located adjacent to the luminosity PCB **54** or the mode selector PCB **84**. These spacers **86**, **88** allow clearance between: (i) moving parts contained within the first end assembly **24** and (ii) the circuit components (e.g., capacitors, resistors, microprocessor, and etc.) that are operably coupled to the PCBs **54**, **84**. Specifically, the first and second spacers **86**, **88** only extend partially around the perimeter of the mode selector PCB **84** to ensure that they do not interfere with the moving parts of the first end assembly **24**. The first and second spacers **86**, **88** also have projections **90** that align with recesses that are formed in interior of the first end assembly **24**. These projection **90** help ensure the proper positioning of the PCBs **54**, **84** and other components within the first end assembly **24**.

(46) The mode selector assembly **32** allows the user to select the operating mode of the portable light **10**. In particular, the mode selector assembly **32** allows the user to select one of the lighting elements **100** for illumination or a plurality of lighting elements **100** for illumination. This selection is performed by angularly displacing the mode selector **32** about the central axis A-A. Referring to FIGS. **9** and **13**, the mode selector assembly **32** includes: (i) the mode selector ring **82**, (ii) the mode selector PCB **84**, (iii) the mode selector aperture **70**, (iv) a mode selector arm **92**, (v) screw **94**, (vi) a spring **96**, and (vii) a ball **98**. The mode selector ring **82** is slidably coupled to the upper housing and circumscribes a portion **102** of the upper housing **26**. In particular, mode selector ring **82** is positioned between the upper flange **78** of the upper housing **26** and the first connector ring

34. The mode selector ring **82** is configured to be angularly displaced relative to the upper housing **26**. As shown in FIGS. **1-5**, the mode selector ring **82** may have a textured or knurled (e.g., strait, angled, or crossed line) engaging surface **106** and a substantially flat, minor outer surface **104**. The textured or knurled surface **106** may be included to enhance a user's grip when rotating the mode selector ring **82**. It should be understood that that the textured or knurled surface **106** may be omitted (see FIGS. **23-25**) or may be reduced in size. The minor outer surface **104** may contain a selection indicator **108** to indicate which illumination mode has been selected.

(47) During operation, the user aligns the selection indicator **108** with an illumination mode indicia **110** that is disposed on the lateral face of the upper flange **78**. Specifically, the mode indicia **110** may include “OFF” **110a**, “I” **110b**, “II” **110c**, “III” **110d**, and “IV” **110e**. The upper flange **78** does not move relative to the mode selector ring **82**. Thus, the application of an angularly force on the mode selector ring **82** relative to the upper flange **78** results in a movement of the selection indicator **108** relative to the mode indicia **110**. It should be noted that the mode selector ring **82** may only need to be angularly displaced approximately 100 degrees to move the mode selector ring **82** from a first or “OFF” position to a final or all on position. Nevertheless, it should be understood that other degrees of angularly displacement (e.g., 5 degrees to 360 degrees) between the operational modes may be chosen.

(48) The mode selector ring **82** also has a substantially smooth inner surface **111** with a recess **112** formed therein. The mode selector arm **92** is coupled to the center of the mode selector PCB **84** by the screw **94** and is configured to extend through the mode selector aperture **70** and into an extent of the recess **112** that is formed within the inner surface **111** of the mode selector ring **82**. A spring **96** and a ball **98** are configured to be positioned between the mode selector arm **92** and the mode selector PCB **84**. The ball **98** is configured to be positioned within dimples **114** formed in the mode selector PCB **84**. During operation, the user may apply an angular force on the mode selector ring **82** to move the selection indicator **108** from the mode indicia **110a**. This clockwise angular force applies pressure on the ball **98**, such that the ball **98** applies a downward pressure on the spring **96**, which allows the ball **98** to move out of a dimple **114a** formed in the mode selector PCB **84**. The ball **98** then slides radially along the mode selector PCB **84**, while staying in contact therewith. Once the user has moved the mode selector ring **82** to the point that the selection indicator **108** is aligned with the mode indicia **110b**, the ball **98** is forced by the spring **96** into a different dimple **114b** formed in the mode selector PCB **84**. Once the ball **98** makes contact with the different dimple **114**, operational mode of the portable light **10** is altered. Additional details about the operation of the mode selector assembly **32** are discussed in greater detail below.

(49) It is to be understood that the mode selector assembly **32** can be replaced with an assembly that is configured to utilize one or more buttons, switches, sliders, local sensors (e.g., motion, light, sound, heat, motion), or other types of electrical selection devices. In one embodiment, the mode selector ring **82** and its associated circuitry may be replaced by a button and the necessary circuitry to allow a user to select the desired operational mode by depressing the button a predetermined number of times. For example, the user: (i) may depress the button one time to turn “On” on lighting element **100**, (ii) may depress the button two time to turn “On” on two of the lighting element **100**, or (iii) may depress the button five time to turn “Off” on lighting element **100**. Additionally, the luminosity selector assembly **28** may be remotely located from the portable light **10**. For example, the luminosity selector assembly **28** may be located on a remote wireless device, such as the remote devices described in U.S. patent application Ser. No. 15/812,852, filed on May 17, 2018, and which is fully incorporated herein by reference. Another example of a remote wireless device includes a cellphone, laptop, RF remote control, or other devices that are connected to the portable light **10** via the internet (e.g., wireless camera, motion sensor, light sensor, timer, etc.).

(50) As best shown in FIG. **9**, the lighting element holder **118** electrically couples the lighting elements **100** to a first power source PCB **124**. The lighting element holder **118** includes

substantially straight sections **120** that are connected to one another by angled projections **122**. The angled projections **122** extend both above and below the substantially straight sections **120**. The top extend of the angled projections **122** define a receiver, which receives an extent of a positioning ring **116**. This configuration centers the lighting element holder **118** in the middle of the portable light **10**. The bottom extent of the angled projections **122** operably couple the first power source PCB **124** to the lighting elements **100** by applying pressure to an extent of the outer surface of the lighting elements **100**. The coupling of the first power source PCB **124** and the lighting elements will be described in greater detail below.

(51) The first power source PCB **124** is configured to electrically couple the lighting elements **100** with: (i) the power source **134**, (ii) the first power source PCB **124**, (iii) the mode selector PCB **84**, (iv) the luminosity PCB **54**, and (v) other circuitry components. Referring to FIG. **9**, the first power source PCB **124** includes a top surface **126** and a bottom surface **128**. The top surface **126** includes surface mounted circuitry, such as resistors, capacitors, or etc. The bottom surface **128** of the first power source PCB **124** includes a power source contact that is made from a conducting material and is configured to make contact with a power source **134**. In particular, the power source contact includes two separate contacts that are coplanar within one another, wherein the first contact **123** is a small circular disk located in the center of the portable light **10** and the second contact **125** is a ring that encircles the first contact **123**. As will be discussed in greater detail below, the first contact **123** is configured to receive a positive charge from the power source **134**, while the second contact **125** is configured to receive a negative charge from the power source **134**. It should be understood that in an alternative embodiment the first contact **123** may receive a negative charge from the power source **134**, while the second contact **125** may receive a positive charge from the power source **134**. In addition, the configuration of the contacts that form the power source contact may include other configurations, such as contacts that are positioned side by side or that are positioned on different heights when compared to one another.

(52) The first power source PCB **124** also includes projections **132** and PCB lighting contacts **130** that are disposed on the sides of the first power source PCB **124**. The projections **132** extend radially outward from the first power source PCB **124** and include apertures **138** formed therethrough. The apertures **138** are configured to receive an extent of the intermediate section **22** of the portable light **10**. The PCB lighting contacts **130** are formed between the projections **132**. This configuration of projections **132** and apertures **138** helps ensure that the PCB lighting contacts **130** are properly aligned with intermediate section **22**. Specifically, the lighting elements **100** have lighting element contacts **129** that are disposed on the rear surface **136** of lighting elements **100**. When the portable light **10** is assembled, the lighting element contacts **129** are positioned adjacent to the PCB lighting contacts **130**. The positional relationship between the lighting element contacts **129** and the PCB lighting contacts **130** is maintained by an inward directed force that is applied to the outer surface of the lighting elements **100** by the lighting element holder **118**. It should be understood that other methods, such as the soldering of wires to each of the contacts, may be used to operably couple the lighting elements **100** to the first power source PCB **124**.

(53) Referring to FIGS. **1**, **9-11**, and **13**, the intermediate section **22** of the portable light **10** includes a middle or intermediate housing **140** and one or more lighting elements **100**. The intermediate housing **140** extends between the first end assembly **24** and a second end assembly **142**. The intermediate housing **140** includes: (i) a central body **144** that has a first end **146** and a second end **148**, (ii) a plurality of external ribs **150** that extend both radially from the central body **144** and longitudinally along a length $L_{sub.CB}$ of the central body **144**, (iii) a plurality of channels **158** that extend laterally between a pair of the ribs **150** and longitudinally along a length $L_{sub.CB}$ of the central body **144**, (iv) a set of projections **151** that extend from a first end **146** of the central body **144**, and (v) a set of recessed sections **152** that are positioned proximate to the second end **148** of the central body **144**. Preferably, the projections **151** are angularly aligned with the ribs **150** (see FIG. **13**), such that a projection **151** extends longitudinally from a rib **150**. The central body

144 has an elongated substantially cylindrical configuration, where the length L.sub.CB of the central body **144** (see FIG. 9) substantially exceeds the width of the central body **144**. It should be understood that in other embodiments, the central body **144** may have an alternative configuration, such as a triangular prism, a rectangular prism, a cube, a pentagonal prism, a hexagonal prism, octagonal prism, sphere, a cone, a tetrahedron, a cuboid, a dodecahedron, a icosahedron, a torus, a octahedron, a ellipsoid, or any other similar shape

(54) The central body **144** is configured to be operably coupled to the first end assembly **24** and the second end assembly **142**. Specifically, exterior threads **154** that are formed in the projections **151** of the central body **144** are configured to be received by the second set of internal threads **74** formed in the upper housing **26**; thereby, operably coupling the first end assembly **24** to the intermediate housing **140**. The exterior threads **196** that are formed in the recessed sections **152** of the central body **144** are configured to be received by a set of internal threads **194** formed in a base connector **184**; thereby, operably coupling the second end assembly **142** to the intermediate housing **140**.

(55) A receiver **153** is formed within the central body **144** and is configured to receive at least a substantial extent of the power source **134**, and preferably the entirety of the power source **134**. In the embodiment of the Figures, including FIGS. 9 and 10, the power source **134** is completely contained within the receiver **153**, wherein the power source **134** is radially inward from: (i) the central body **144**, (ii) the plurality of external ribs **150**, (iii) a plurality of channels that are formed between the ribs **150**, and (iv) the lighting elements **100**. In other words, the receiver **153** is configured to surround the power source **134**, which in turn positions the power source **134** substantially in the center of the portable light **10**.

(56) The plurality of external ribs **150** extend radially outward from the central body **144** and longitudinally along a length L_{es} of the central body **144**. As shown in FIG. 10, the central body **144** has both an inner diameter and an outer diameter that are less than an outer diameter defined by an outer surface **176** of the plurality of ribs **150**. Each rib **150** includes two lateral projections **160** that extend in opposite lateral directions from one another, which provides the rib **150** with a “T-shape” in cross section. Like the ribs **150**, the lateral projections **160** also extend longitudinally along a length L.sub.CB of the central body **144**. The lateral projections **160** form a recess **161** on each side of the rib **150**, that as discussed below is configured to receive an extent of the lens **168**. In one embodiment (see FIGS. 10-11), the ribs **150** may be evenly spaced a distance apart circumferentially around the central body **144**. Accordingly, this configuration alternates between ribs **150** and lenses **168** to provide a portable light **10** that can radiate light omni-directionally or 360 degrees. In alternative embodiments, the ribs **150** may not be evenly spaced around the central body **144**.

(57) The area between each pair of ribs **150** forms a channel **158**. Accordingly, each channel **158** extends longitudinally along the central body **144**. These channels **158** can have an arced or curvilinear configuration, as shown in FIGS. 10 and 11. Alternatively, the channels **158** may be substantially flat, as shown in FIG. 13. Each channel **158** is configured to receive one of the lighting elements **100**. This configuration allows for the lighting elements **100** to be positioned radially outward of the central axis A-A, while extending longitudinally along the housing **140** in a direction that is parallel to, or substantially parallel, to the central axis A-A. This in turn allows light that is emitted from the one or more lighting elements **100** to primarily radiate in a direction that is perpendicular to a longitudinal axis of the housing **140** and a longitudinal axis of the power source **134**. In other words, the light that is emitted from the one or more lighting elements **100** is not configured to emit light that is primarily parallel with the longitudinal axis of housing **140** or the longitudinal axis of power source **134**.

(58) As mentioned above, each channel **158** is configured to receive one of the lighting elements **100**. Accordingly, the lighting elements **100** are configured to extend longitudinally along the central body **144**. As shown in FIGS. 9 and 13, the lighting elements **100** extend across almost the

entire length of the power source **134**. In other words, the lighting elements **100** have a length that is nearly (e.g., 90% of the length) as long as the length of the power supply **134**. Also, as shown in FIGS. **9** and **13**, the lighting elements **100** have a length that is greater than the central body **144**. This configuration facilitates the retention of the lighting elements **100** within the portable light **10**. (59) Each lighting elements **100** has a lighting element PCB **162** and an emitter assembly **164**. The lighting element PCB **162** is disposed radially inward from the emitter assembly **164** and the lens **168** and is configured to reside within the channels **158**. As shown in FIGS. **9** and **13**, the lighting element PCB **162** has a longer configuration than the emitter assembly **164** and the lens **168**, which allows the lighting element PCB **162** to extend into an extent of the first and second assemblies **24**, **142** and allows the emitter assembly **164** to reside outside of the first and second assemblies **24**, **142**. This configuration facilitates the operable coupling of the power source **134** and the emitter assembly **164**. Accordingly, the lighting element PCB **162** extends longitudinally along the central body **144**.

(60) As mentioned above, the lighting elements **100** have lighting element contacts **129** and a rear surface **136**. The specific structure of the lighting elements **100** that contain these structures is the lighting element PCB **162**. Thus, the lighting element PCB **162** is configured to be operably couple to the first power source PCB **124**. The emitter assembly **164** is also configured to be operably coupled to the emitter assembly **164**. Accordingly, the lighting element PCB **162** allows current to flow from the power source **134** to the emitter assembly **164** when the portable light **10** is in the "On" position.

(61) The emitter assembly **164** is disposed on the outer surface of the lighting element PCB **162**, while being positioned radially inward from the lens **168**. Each emitter assembly **164** may be composed of between 10 and 200 individual emitters **165**, preferably between 15 and 150 individual emitters **165**, and most preferably between 50 and 100 individual emitters **165**. Every individual emitter **165**, which is a part of the emitter assembly **164**, are configured to illuminate when power is applied to the emitter assembly **164** by the lighting element PCB **162**. Each emitter assembly **164** may produce between 0 and 1000 lumen, preferably between 0 and 750, and most preferably between 0 and 500. Accordingly, a portable light **10** that has four light emitter assemblies **164** may produce between 0 and 4000 lumen, preferably between 0 and 3000, and most preferably between 0 and 2000. It should be understood that is other embodiments every individual emitter **165** contained within the emitter assembly **164** may not be configured to illuminate when power is applied to the emitter assembly **164**. For example, a user may be able to select the desired individual emitters **165** that the user wants illuminated, while keeping other individual emitters **165** in the unilluminated.

(62) As shown in FIGS. **9-11**, the portable light **10** does not include a secondary optic nor does it include a reflector. Accordingly, substantially all of the light that is emitted from the emitter assembly **164** passes directly from the emitter assembly **164**, through the lens **168**, and to the surrounding environment. This configuration may be desired, as substantially no light is lost due to material absorptions outside of the lens **168**. Further, this configuration allows for a uniform and wide distribution of light. This configuration may be accomplished by using a Chip on Board ("COB") LED or surface-mount device LED. In this configuration there is no secondary optic and the lens **168**, which acts as the primary optic, can be configured to overlay multiple individual light emitters **165** that are a part of the emitter assembly **164**. Specifically, if a COB LED is used as the emitter assembly **164**, then the lens **168** may overlay at least 20 individual light emitters **165** and preferably all of the individual light emitters **165** (e.g., 60 individual light emitters).

(63) In other embodiments, a secondary optic or a reflector may be used. For example, if a standard LED is selected as the emitter, the portable light **10** may have a secondary optic. In this configuration, the primary optic for the standard LED is the optic that surrounds the LED, while the secondary optic is the lens **168**. This secondary optic may be configured to protect the standard LEDs from the surrounding environment and protects the user from contacting the hot outer surface

of the primary optics after the LEDs. Alternatively, the portable light **10** may include a reflector that is positioned between the edges of the emitter assembly **164** and the ribs **150**. This reflector may help focus the light that is emitted from the emitter assembly **164** in a specific direction to achieve a desired light distribution. In a further alternative, the emitter assembly **164** may be positioned such that it is facing inward, towards the center of the portable light **10**, and the reflector may direct light from the emitter assembly **164** out through the lens **168**. This configuration may provide a softer lighting effect, as no direct light that is emitted from the emitter assembly **164** can exit the portable light **10** without being reflected. It should be understood that any combination of these alternative embodiments may be combined in the portable light **10**. For example, the elongated light source may have: i) lighting element **100** that has a recessed emitter assembly **164**, which includes reflectors that extends from the outer edges of the recessed emitter assembly **164** to the outer edges of the ribs **150**, and ii) lighting elements **100** that are positioned on either side of the first lighting element **100** that do not have a reflector or a secondary optic.

(64) The lens **168** is positioned radially outward from the lighting element PCB **162** and the emitter assembly **164** and functions as a primary optic for the lighting element **100**. The lens **168** has an inner surface **172** and an outer surface **174**, where the outer surface **174** is configured to be radially aligned with an outer surface **176** of the ribs **150**. This configuration provides a substantially smooth outer surface of the portable light **10**. The lens **168** includes one or more lens tabs or projections **170**, which are dimensioned to be received by the recess **161** to secure the peripheral edges of the lens **168** under the lateral projections **160** of the ribs **150**. The lens **168** serves to physically protect the lighting element PCB **162** and the emitter assembly **164**, while allowing at least a substantial extent or entirety of the light generated by the emitter assembly **164** to pass through the lens **168**. As shown in FIG. **10**, the central body **144** has both an inner diameter and an outer diameter that are less than an outer diameter defined by an outer surface **174** of the plurality of lenses **168**. The outer surface **174** of each lens **168** has a curvilinear configuration and the outer surface **176** of each rib **150** has a curvilinear configuration, wherein these outer surfaces **174**, **176** are substantially flush with each other. Also, the outer surfaces **174**, **176** combine to provide a substantially smooth outer curvilinear surface for the central body **144** that is devoid of appreciable gaps or ridges, which helps the user to easily grasp the central body **144** for securement of the light **10** and improves the aesthetic appearance of the light **10**.

(65) Depending on the type of selected emitter assembly **164** and the configuration of the channels **158**, the inner surface **172** of the lens **168** may be in contact with an outer surface of the emitter assembly **164** (see FIGS. **10-11**) or there may be an air gap **166** that is positioned between the outer surface of the emitter assembly **164** and the inner surface **172** of the lens **168**. This air gap **166** may be filled with standard air, an inert gas, or a material that changes the reflective properties of the emitter assembly **164** (e.g., fused quartz or fluorite). In an alternative embodiment, the lens **168** may be omitted, its configuration may be changed, or there may be multiple lenses. For example, lens **168** may be omitted if the emitter assembly **164** that is chosen already has a protective cover.

(66) In some implementations, the intermediate section **22** includes four lighting elements **100a**, **100b**, **100c**, and **100d** that are positioned radially around the housing **140** and the power source **134**. The four lighting elements **100a**, **100b**, **100c**, and **100d** may be evenly spaced radially around the exterior of the intermediate section **22**. This configuration allows for each lighting element **100** to project light in a 90 degree radial pattern around the portable light **10**. Accordingly, when two lighting elements **100** are illuminated, the portable light projects light in a 180 degree pattern around the portable light **10**. Similarly, when all four lighting elements **100** are illuminated, light is projected in a 360 degree pattern around the portable light **10**. This configuration positioned the power source **134** between lighting elements **100a** and **100c**. Likewise, the power source **134** is also positioned between lighting elements **100b** and **100d**. Specifically, the power source **134** is centered between lighting elements **100a** and **100c**. Likewise, the power source **134** is centered between lighting elements **100b** and **100d**. Accordingly, the lighting elements **100** are configured to

project light in a 360 degree pattern around the power source **134**.

(67) It should be understood that the lighting elements may **100a**, **100b**, **100c**, and **100d** not be evenly spaced around the exterior of the intermediate section **22**. For example, it may be beneficial to place two lighting elements abutting one another to provide a larger quantity of focused light in one area. Additionally, it should be understood that more or less than four lighting elements **100** may be included in the intermediate section **22**. For example, the intermediate section **22** may include one lighting element **100** or ten lighting elements **100** that extend the entire length of the intermediate section **22**. Specifically, one lighting element **100** may be utilized that encircles the entire central body **144**. Further, the other configurations of lighting elements **100** are contemplated herein. For example, the intermediate section **22** may include eight lighting elements **100**, where each lighting element **100** does not extend the entire length of the intermediate section **22**. Instead, each lighting elements **100** only extends from the edge of the intermediate section **22** to the middle of the intermediate section **22**. It should be understood that any combination of the above embodiments is contemplated by this disclosure.

(68) The portable light **10** further has multiple mounting configurations for use in different environments and under different conditions. For example, the user can move the securement member **30** to a deployed position **216**, as best shown in FIGS. **1** and **2**, where upon the user can then hang the portable light **10** via the deployed securement member **30** to an object or fixture (e.g., a hook) disposed above the portable light **10**. Second, the user may releasably attach the magnet **190** residing within the lower end **18** to a magnetic surface or object. Thus, the portable light **10** will be magnetically mounted, to a selected magnetic surface or object at any orientation (e.g., to a tool box or to a metal surface of a vehicle). Third, the user may place the bottom surface **199** of the portable light **10** on a surface support surface **44** and deploy a retractable stabilization assembly **192** (see FIG. **2**). Fourth, the user may manually carry the portable light **10** for portable illumination use as a flashlight or lantern between various locations.

(69) The plurality of ribs **265** that are formed in the battery cartridge **250** may include three elongated ribs that connect the first end cap **252** to the second end cap **254**. These elongated ribs **265** hold the removable non-rechargeable batteries **278** (e.g., 9 AA batteries) within the battery cartridge **250**. To facilitate the holding of the batteries **278** within the battery cartridge **250**, the ribs **265** may have projections that are configured to extend around a portion of the batteries **278**, when the batteries **278** are placed within the battery cartridge **250**. It should be understood that different configurations of a battery cartridge **250** may be implemented, such that additional (e.g., 12 or 15 total batteries) or fewer (e.g., 1-9 total batteries) batteries **278** are used or larger (e.g., C or D) or smaller (e.g., AAAA or AAA) battery sizes.

(70) Positioned between the elongated ribs **265** and the end caps **252**, **254** are the first and second battery cartridge PCBs **260**, **262**, respectively. The first battery cartridge PCB **260** includes a plurality of contacts **280** disposed on an inner surface and a plurality of contacts **282** disposed on an outer surface. The inner surface contacts **280** are configured to make contact with the positive charged end of the batteries **278**, while the outer surface contacts **282** are configured to make contact with the first plurality of end cap contacts **256**. The first battery cartridge PCB **260** electrically couples together all of the inner surface contacts **280** with end cap contact **262**. This enables the positive charge from all of the removable non-rechargeable batteries **278** to be connected to the first contact **123** on the first power source PCB **124**, when the power source **134** is positioned within the portable light **10**. The first battery cartridge PCB **260** electrically couples together the outer surface contacts **282** that are electrically connected to end cap contacts **260** and **264**.

(71) The second battery cartridge PCB **262** includes a plurality of contacts **284** disposed on an inner surface and a plurality of contacts **286** disposed on an outer surface. The inner surface contacts **284** are configured to make contact with the negative charged end of the batteries **278**, while the outer surface contacts **286** are configured to make contact with the second plurality of

end cap contacts **258**. The second battery cartridge PCB **262** electrically couples together all of the inner surface contacts **284** with end cap contacts **266** and **270**. This enables the negative charge from all of the removable non-rechargeable batteries **278** to be connected to the second power source PCB **178**, when the power source **134** is disposed within the portable light **10**. The connecting wire **264** extends from the first battery cartridge PCB **260** to the second battery cartridge PCB **262** and electrically couples end cap contacts **260** and **264** with contact **268**.

(72) Referring to FIGS. **26** and **27**, the configuration of the power source **134** allows it to supply power to the portable light **10** regardless of which direction the power source **134** is inserted within the portable light **10**. In other words, the portable light **10** will work when first end cap **252** is placed adjacent to the first power source PCB **124** or when the first end cap **252** is placed adjacent to the second power source PCB **178**. The following describes how current flows through the portable light **10**, when the first end cap **252** is placed adjacent to the first power source PCB **124** (see FIG. **27**). The positive current flows from the removable non-rechargeable batteries **278** to all of the inner surface contacts **280** on the first battery cartridge PCB **260**. Next, the current flows from the inner surface contacts **280** to the outer surface contact **282** that correspond to end cap contact **268**. Next, current flows from the outer surface contact **282** that correspond to end cap contact **268** to end cap contact **268**. Next, current flows from end cap contact **268** to the first contact **123** associated with the first power source PCB **124**. Next, the current flows from the first contact **123** associated with the first power source PCB **124** to the circuitry **499** contained within the portable light **10**, including the lighting elements **100**. Next, the current flows from the circuitry **499** contained within the portable light **10** to the second contact **125** associated with the first power source PCB **124**. Next, the current flows from second contact **125** associated with the first power source PCB **124** to end cap contacts **266** and **270**. Next, the current flows from the end cap contacts **266** and **270** to outer surface contacts **282** that are associated with end cap contacts **266** and **270**. Next, the current flows from the outer surface contacts that are associated with end cap contacts **266** and **270** to the connecting wire **264**. Next, the current flows from the first end of the connecting wire **264** to the second end of the connecting wire **264**. Next, the current flows from the second end of the connecting wire **264** to the outer surface contact **286** that is associated with end cap contact **274**. Next, the current flows from outer surface **286** contact that is associated with end cap contact **274** to end cap contact **274**. Next, the current flows from the end cap contact **274** to the second power source PCB **178**. Next, the current flows from the second power source PCB **178** to the end cap contacts **272** and **276**. Next, the current flows from the end cap contacts **272** and **276** to the outer surface contacts **286** of the second battery cartridge PCB **262** to all of the inner surface contact **284**. Finally, the current flows from the inner surface contacts **284** to the negative charged ends of the batteries **278**.

(73) Alternatively, the following describes how current flows through the portable light **10**, when the second end cap **254** is placed adjacent to the first power source PCB **124** (see FIG. **28**). The negative current flows from the removable non-rechargeable batteries **278** to all of the inner surface contacts **284** on the second battery cartridge PCB **262**. Next, the current flows from the inner surface contacts **284** to the outer surface contact **286** that correspond to end cap contacts **272** and **276**. Next, current flows from the outer surface contacts **286** that correspond to end cap contacts **272** and **276** to end cap contacts **272** and **276**. Next, current flows from end cap contacts **272** and **276** to the second contact **125** associated with the first power source PCB **124**. Next, the current flows from the second contact **125** associated with the first power source PCB **124** through the circuitry **499** contained within the portable light **10**, including the lighting elements **100**. Next, the current flows from the circuitry **499** contained within the portable light **10** to the first contact **123** associated with the first power source PCB **124**. Next, the current flows from first contact **123** associated with the first power source PCB **124** to end cap contact **274**. Next, the current flows from the end cap contact **274** to outer surface contacts **286** that is associated with end cap contact **274**. Next, the current flows from the outer surface contacts **286** that is associated with end cap

contact **274** to the connecting wire **264**. Next, the current flows from the first end of the connecting wire **264** to the second end of the connecting wire **264**. Next, the current flows from the second end of the connecting wire **264** to the outer surface contact **282** that is associated with end cap contacts **266** and **270**. Next, the current flows from the end cap contacts **266** and **270** to the second power source PCB **178**. Next, the current flows from the second power source PCB **178** to the end cap contact **268**. Next, the current flows from the end cap contact **268** to outer surface contact **282** that is associated with end cap contact **268**. Next, the current flows from the outer surface contact **282** that is associated with end cap contact **268** to all of the inner surface contacts **280**. Finally, the positive current flows from the inner surface contacts **280** to the positive charged ends of the batteries **278**. Accordingly, the power source **134** can be operably connected to the portable light **10** in either direction.

(74) It should be understood that instead of being a combination of removable non-rechargeable batteries **278** disposed within a battery cartridge **250**, the power source **134** may be a removable non-rechargeable battery, a removable rechargeable battery, a combination of removable rechargeable batteries, a combination of removable rechargeable batteries disposed within a battery cartridge **250**, a non-removable rechargeable battery, a combination of non-removable rechargeable batteries disposed within a battery cartridge **250**, a DC power supply that is configured to connect to a 12 volt car battery, a DC power supply that is configured to connect to a 110 volt alternative current outlet, or any other type of power supply that is known to a person of skill in the art.

(75) The second end assembly **142** is operably connected to the second end **18** of the portable light **10**. The second end assembly **142** allows the portable light to be mounted in a multitude of different positions (e.g., one a magnetic support surface **44**, with the retractable stabilization assembly **192** in a retracted state **210**, or with the retractable stabilization assembly **192** in a deployed state **212**). The second end assembly **142** includes: (i) a connector ring **182**, (ii) base connector **184**, (iii) a bottom housing **186**, and (vi) retractable stabilization assembly **192**. The base connector **184** has a first set of internal threads **194** and two sets of external threads **196**, **198**. The first set of internal threads **194** are configured to engage the set of threads **196** that are positioned within the set of recessed sections **152** that are positioned proximate to the second end **148** of the central body **144**. This configuration secures the lighting elements **100** within the channels **158** that are formed within the intermediate housing **140**. While the first set of external threads **196** are configured to engage with the threads on the connector ring **182**, the second set of external threads **198** are configured to operably engage with a set of internal threads **202** positioned within the bottom housing **186**. The threads **198**, **202** enable a user to remove the bottom housing **186** from the intermediate housing **140**. This in turn permits a user to replace the power source **134** or couple additional accessories to the portable light **10**. It should be understood that the bottom housing **186** may be coupled to the intermediate housing **140** using other methods, such as quarter turn connector, a bayonet connector, a pressure fit connector, or other connectors known to one of skill in the art.

(76) The bottom housing **186** includes: (i) the second power source PCB **178**, (ii) a magnet receiver **188**, and (iii) a magnet **190**. The bottom housing **186** may have a textured or knurled (e.g., strait, angled, or crossed line) outer surface **187**. The textured or knurled surface **187** may be included to enhance a user's grip when rotating the bottom housing **186**. It should be understood that the textured or knurled surface **187** may be omitted (see FIGS. 23-25) or may be reduced in size. As discussed above the second power source PCB **178** is configured to make contact with the power source **134** to complete the electrical circuit. The magnet receiver **188** extends from the bottom surface **199** of the bottom housing **186**. This configuration places the magnet **190** a safe distance away from the second power source PCB **178** to ensure that the magnet does not interfere with the electrical circuitry of the portable light **10**. The magnet **190** is disposed within the magnet receive **188** and is substantially flush, or in a common plane, with the bottom surface **199** of the bottom housing **186**. This configuration ensure places the largest surface area of the magnet **190** in contact

with the support surface **44**. The strength of the magnet is sufficient to hold the portable light **10** in a horizontal position from a vertical support surface **44**. Specifically, the strength of the magnet is between 300 and 30 millitesla, preferably between 200 and 75 millitesla, and most preferably between 150 and 100 millitesla. Alternatively, the magnet **190** may be weaker, substantially stronger, or may be electromagnetically controlled by a battery.

(77) The retractable stabilization assembly **192** is shown FIGS. **1-5**, **7-9**, **11**, **13**, and **23-25**. In some implementations, as best shown in FIGS. **2**, **3**, **7-9**, the retractable stabilization assembly **192** includes one or more feet **204**. The feet **204** are individually retractable and can be selectively positioned into an extended position **212**, as exemplarily shown in FIGS. **2** and **8**, and can also be selectively positioned into a retracted position **210** as exemplarily shown in FIGS. **1**, **3-5**, and **7**. When the feet **204** are in the retracted position **210**, the outer surface of the feet **204** do not extend outside the radius of the outer surface of the ribs **150**. When the feet are in the extend position **212**, the outer surface of the feet **204** extend outside the radius of the ribs **150**. This configuration enables the portable light **10** to more stably stand on the support surface **44**, as shown in FIG. **2**.

(78) The bottom housing **186** includes a recess **206** that is formed, at least partially by, the magnet receiver **188**. The feet **204**, or at least a portion of the feet **204**, form a substantially flat and flush surface, or a common plane, with the bottom surface **199** of the bottom housing **186**. The feet **204** can also form a common plane with a magnet **190**. Additionally, the feet **204**, or at least a portion of the feet, can be disposed within the recess **206**, when the feet **204** are disposed in the retracted position **210**. This configuration is facilitated by the substantially arcuate shape of the feet **204**. Specifically, when the feet **204** are disposed in the retracted position **210**, the feet **204** can form substantially a circle, and thus conform to the generally cylindrical shape of the portable light **10**.

(79) Each of the one or more feet **204** can be hingedly attached to the bottom housing **186** of the portable light **10** at a pivot point **208**, which can be formed a fastener **209**. The feet **204** can be biased towards one or more of the extended position **212** and the retracted position **210**. The feet **204** can also lock, through a releasable locking system (not shown) in one or more of the extended position **212** and the retracted position **210** by any mechanical fastening system commonly known to those skilled in the art.

(80) Referring to FIGS. **1-12**, the securement member **30** can support, suspend and/or brace the portable light **10** during operation, storage and/or transportation. The first end **14** includes a securement member **30**, which is pivotally attached to the first end **14**. Specifically, the securement member **30** is coupled to the upper housing **26** at a pivot point **31**, which is disposed on the outer surface of the upper flange **78**. This configuration allows the securement member **30** to be pivotally attached to the first end **14** and can selectively rotate among a plurality of orientations with respect to the first end **14**.

(81) The securement member **30** may have a substantially arcuate shape, but it is to be understood that other shapes are within the scope of this disclosure. The securement member **30** may be selectively disposed in a first, or stowed, position **214**, as exemplarily shown in FIGS. **3-6**, **8-9**, and **11**. In the first position **214**, the securement member **30** is positioned between the upper edge of the mode selector ring **82** and the outer rim **40**. Thus, the securement member **30** is positioned adjacent to the outer surface of the upper flange **78**. In this first position **214**, the securement member **30** lies in a plane substantially parallel with the first end wall **38** and the outer rim **40**.

(82) The securement member **30** may also be selectively disposed in a second, or deployed, position **216**, as exemplarily shown in FIGS. **1-2**. In the second position **216**, the securement member **30** is positioned above the outer rim **40**. In this second position **216**, the securement member **30** may be positioned at any angle that places it above the outer rim **40** and is fully deployed when securement member **30** is positioned substantially perpendicular to the outer rim **40**. It should be understood that the securement member **30** may also have a releasable locking system (not shown), which may secure the securement member **30** in one or more positions (e.g., first position **214** or the second position **216**).

(83) FIG. 12 is a circuit diagram of the components that are contained within the portable light 10. In particular, the circuit components shown in FIG. 12 include: i) switches 218-226 that correspond to mode indicia 110a-110c, ii) switch 228 that corresponds to the luminosity selector assembly 28, iii) the power source 134, iv) lighting elements 100a-100d, v) resistors and capacitors that are disposed either the luminosity PCB 54, the mode selector PCB 84, or the first power source PCB 124, which include R5-R8, R12, R13, R15, C1-C3, and vi) microcontroller 230. In this configuration, the microcontroller 230 accepts inputs from the switches 218-228 and power from the power source 134 and outputs control signals to the lighting elements 100a-110d. The circuit elements that are disposed on the either of the PCBs 54, 84, or 124 protect the microcontroller from current spikes or electrical noise, act as an electrical filter, or act as a dimming potentiometer for the luminosity selector assembly 28. Also, as shown in FIG. 12, each lighting element 100a-100d includes: i) emitter assembly 164, ii) two resistors (e.g., R10 and R14), iii) and a transistor (e.g., Q4). Each of these circuit elements contained within the lighting elements 100a-100d are disposed on the lighting element PCB 162. It should be understood that other circuit configuration or the placement of circuit elements on other PCBs is contemplated by this disclosure. Further, it is contemplated by this disclosure that other circuit elements may be included within the portable light 10, such as a wireless communication module, additional microcontrollers, additional switches to control the lumen output or the selection of emitters 164, or any other circuitry required to perform the functions described herein.

(84) The below disclosure of the operation of portable light 10 applies in equal force to all embodiments (e.g., the first embodiment shown in FIGS. 1-13, the second embodiment shown in FIGS. 14-22, and the third embodiment shown in FIGS. 23-25) of the portable elongated lights described herein. Specifically, in operation, a user can: (i) selectively mount the portable light 10 in various ways, (ii) select the luminosity level of the lighting elements 100, and (iii) select which lighting elements are illuminated 100. For example, the user can move the securement member 30 to a deployed position 214, as best shown in FIGS. 1 and 2, where upon the user can then hang the portable light 10 via the deployed securement member 30 to an object or fixture (e.g., a hook) disposed above the portable light 10. Second, the user may releasably attach the magnet 190 residing within the lower end 18 to a magnetic surface or object. Thus, the portable light 10 will be magnetically mounted, to a selected magnetic surface or object at any orientation (e.g., to a tool box or to a metal surface of a vehicle). Third, the user may place the bottom surface 199 of the portable light 10 on a surface support surface 44 and deploy a retractable stabilization assembly 192 (see FIG. 2). Fourth, the user may manually carry the portable light 10 for portable illumination use as a flashlight or lantern between various locations.

(85) The user also has a plurality of options regarding portable light 10 luminosity levels. The luminosity selector assembly 28 may be manipulated to cause the lighting elements 100 of the portable light 10 to produce zero luminosity. Alternatively, the luminosity selector assembly 28 may be manipulated to cause one or more of the lighting elements 100 to produce various degrees of light. While FIGS. 1-11, 14-20, and 23-25 illustrate the lighting elements 100 as producing zero light, FIGS. 21-22 show each of the lighting elements 100 as emitting light, as indicated by light lines 232.

(86) The user further has a plurality of options regarding portable light 10 operational modes. The operational modes can be selected or changed by manipulating the mode selector ring 82 relative to the mode indicia 110. In some implementations, as shown in the figures, the selection indicator 108 on the mode selector ring 82 is rotated relative to the mode indicia 110. The mode indicia 110 may include images or numerals, exemplarily shown as I, II, III and IV. When the selection indicator 108 is arranged correspondingly to one of the mode indicia 110 (I, II, III and IV), the portable light 10 will operate in the operational mode corresponding to the selected mode indicia 110 (I, II, III and IV). It is to be understood that more, fewer or alternate mode indicia 110 are within the scope of this disclosure.

(87) Some of the operational modes are discussed below. In an “OFF” operational mode, the no lighting elements **100** are illuminate. The portable light **10** can be set to this “OFF” operation mode in two different way. A first way this “OFF” operational mode may be achieved is by the user placing the mode selector ring **82** in the “OFF” position by aligning the selection indicator **108** with mode indicia **110a**. In this configuration, it does not matter the setting of the luminosity selector assembly **28** because there are no lighting elements **100** that are selected for illumination. A second way this “OFF” operational mode may be achieved is by the user placing the rotary dial **46** in the “OFF” position **64**. In this configuration, it does not matter the setting of the mode selector assembly **32** because there is no current being applied to any of the lighting elements **100**.

(88) In a first operational mode, the one of the lighting elements **100** is illuminate. For example, lighting element **100a** may be illuminated. This second operation mode is achieved by both: i) rotating the luminosity selector ring **82** to align the selection indicator **108** with mode indicia **110b** and ii) rotating the rotary dial **46** to command greater than zero luminosity. If either one of these assemblies **28**, **32** are positioned in a different location, then the portable light **10** will not be in a second operation mode. It should be understood that the user may then vary the lumen output of the illuminated lighting element (e.g., **100a**) by changing the position of the rotary dial **46**. For example, the user may rotate the dial **46** clockwise to increase the lumen output or may rotate the dial **46** counter-clockwise to decrease the lumen output.

(89) In a second operational mode, the two of the lighting elements **100** are illuminate. For example, lighting element **100a** and **100b** may be illuminated. This third operation mode is achieved by both: i) rotating the luminosity selector ring **82** to align the selection indicator **108** with mode indicia **110c** and ii) rotating the rotary dial **46** to command greater than zero luminosity. If either one of these assemblies **28**, **32** are positioned in a different location, then the portable light **10** will not be in a third operation mode. It should be understood that the user may then vary the lumen output of the illuminated lighting elements (e.g., **100a** and **100b**) by changing the position of the rotary dial **46**.

(90) In a third operational mode, the three of the lighting elements **100** are illuminate. For example, lighting element **100a**, **100b**, and **100c** may be illuminated. This fourth operation mode is achieved by both: i) rotating the luminosity selector ring **82** to align the selection indicator **108** with mode indicia **110d** and ii) rotating the rotary dial **46** to command greater than zero luminosity. If either one of these assemblies **28**, **32** are positioned in a different location, then the portable light **10** will not be in a fourth operation mode. It should be understood that the user may then vary the lumen output of the illuminated lighting elements (e.g., **100a**, **100b**, and **100c**) by changing the position of the rotary dial **46**.

(91) In a fourth operational mode, all four of the lighting elements **100** are illuminate. For example, lighting element **100a**, **100b**, **100c**, and **100d** may be illuminated. This fourth operation mode is achieved by both: i) rotating the luminosity selector ring **82** to align the selection indicator **108** with mode indicia **110e** and ii) rotating the rotary dial **46** to command greater than zero luminosity. If either one of these assemblies **28**, **32** are positioned in a different location, then the portable light **10** will not be in a fifth operation mode. It should be understood that the user may then vary the lumen output of the illuminated lighting elements (e.g., **100a**, **100b**, **100c**, and **100d**) by changing the position of the rotary dial **46**.

(92) In some embodiments, adjacent lighting elements **100**, such as **100a** and **100b** or **100a-100c**, may illuminate when a user manipulates both assemblies **28**, **32** to command such light output. In other embodiments, non-adjacent lighting elements **100**, such as **100a** and **100c** or **100b** and **100d**, may illuminate when a user manipulates both assemblies **28**, **32** to command such light output. Other operational modes may be configured within the portable light **10**. For example, such operational modes may include: i) one or more of the lighting elements **100** operating in a flashing pattern (e.g., strobe, SOS, or etc.), ii) one or more of the lighting elements **100** illuminating a color different in comparison to the other lighting elements **100**, iii) one or more of the lighting elements

100 illuminating different lumen outputs in comparison to the other lighting elements **100** (e.g., **100a** outputs 2000 lumen, while **100b** outputs 500 lumen), iv) one or more of the lighting elements **100** illuminating only part of the lighting element **100** (e.g., one the bottom half of the lighting element **100** is illuminated), v) or a combination of any of these operational modes.

(93) Similar to the portable light **10** as described above, FIGS. **14-22** show another embodiment of a portable light **1001**. The portable light **1001** can be selectively mounted in a various ways, can selected which lighting elements **1100** are illuminated, and select the luminosity level of the lighting elements **1100**. For sake of brevity, the above disclosure in connection with portable light **10** will not be repeated below, but it should be understood that across embodiments like numbers represent like structures. For example, the disclosure relating to lighting elements **100** applies in equal force to lighting elements **1100**. Further, it should be understood that the operational modes of the portable light **1001** are similar to, or identical to, those disclosed regarding portable light **10**. Moreover, it is to be understood that any one or more features of the portable light **10** can be used in conjunction with those disclosed regarding the portable light **1001**, and that any one or more features of the portable light **1001** can be used in conjunction with those disclosed regarding the portable light **10**.

(94) The primary differences between portable light **10** and portable light **1001** are: i) the omission of the securement member **30** from the top assembly **1024**, ii) the replacement of the retractable stabilization assembly **192** with a securement assembly **1600**, and iii) a rotary dial **1046** that has dial grooves **61**. Other than these differences, the portable light **1001** is mechanically and electrically the same as the portable light **10**.

(95) The securement assembly **1600** can support, suspend and/or brace the portable light **10** during operation, storage and/or transportation. The second end **1018** includes a securement assembly **1600**, which is pivotally attached to the second end **1018**. In other words, this configuration allows the securement assembly **1600** to be pivotally attached to the second end **1018** and can selectively rotate among a plurality of orientations with respect to the second end **1018**.

(96) The securement assembly **1600** has a first arcuate member **1602**. The first arcuate member **1602** may be selectively disposed in a first, or stowed, position **1610**, as exemplarily shown in FIGS. **16-18** and **21-22**. In the first position **1610**, the first arcuate member **1602** is positioned adjacent to the magnet **1190** and above the bottom surface **199** of the bottom housing **186**. Thus, the first arcuate member **1602** is positioned within the recess or circular groove **1206**. In this first position **1610**, the first arcuate member **1602** lies in a plane substantially parallel with the bottom surface **199** of the bottom housing **186**. In this second position **1612**, the first arcuate member **1602** may be positioned at any angle that places it below the bottom surface **199** and is fully deployed when first arcuate member **1602** is positioned substantially perpendicular to the bottom surface **199**. It should be understood that the first arcuate member **1602** may also have a releasable locking system (not shown), which may secure the first arcuate member **1602** in one or more positions (e.g., first position **1610** or the second position **1612**).

(97) The securement assembly **1600** may also have a second arcuate member **1604**. The second arcuate member **1604** may be selectively disposed in a first, or stowed, position **1610**, as exemplarily shown in FIGS. **16-18** and **21-22**. In the first position **1610**, the second arcuate member **1604** is positioned adjacent to the magnet **1190** and above the bottom surface **199** of the bottom housing **186**. Thus, the second arcuate member **1604** is positioned within the recess or circular groove **1206**. In this first position **1610**, the second arcuate member **1604** lies in a plane substantially parallel with the bottom surface **199** of the bottom housing **186**. In this second position **1612**, the second arcuate member **1604** may be positioned at any angle that places it below the bottom surface **199** and is fully deployed when second arcuate member **1604** is positioned substantially perpendicular to the bottom surface **199**. It should be understood that the second arcuate member **1604** may also have a releasable locking system (not shown), which may secure the second arcuate member **1604** in one or more positions (e.g., first position **1610** or the second

position **1612**).

(98) When both the first and second arcuate members **1602**, **1604** are in the second position (see FIG. **20**), the arcuate members **1602**, **1604** form an arch that extends from one side of the portable light **1001** to the other side of the portable light **1001**. Additionally, it should be understood that the first and second arcuate members **1602**, **1604** can be moved independently from each other. Further, it is contemplated by this disclosure that the arcuate members **1602**, **1604** may have other shapes or thickness.

(99) Similar to the portable light **10** as described above, FIGS. **23-25** show another embodiment of a portable elongated light **2001**. The portable elongated light **2001** can be selectively mounted in a various ways, can selected which lighting elements **2100** are illuminated, and select the luminosity level of the lighting elements **2100**. For sake of brevity, the above disclosure in connection with portable light **10** will not be repeated below, but it should be understood that across embodiments like numbers represent like structures. For example, the disclosure relating to lighting elements **100** applies in equal force to lighting elements **2100**. Further, it should be understood that the operational modes of the portable elongated light **2001** are similar to, or identical to, those disclosed regarding portable light **10**. Moreover, it is to be understood that any one or more features of the portable light **10** can be used in conjunction with those disclosed regarding the portable elongated light **2001**, and that any one or more features of the portable elongated light **2001** can be used in conjunction with those disclosed regarding the portable light **10**.

(100) The primary differences between portable light **10** and portable elongated light **2001** are: i) the omission of the textured or knurled surface **106** from the mode selector ring **1082**, ii) the omission of the textured or knurled surface **187** from the bottom housing **1186**, and iii) the movement of the securement member **1030** from being positioned adjacent to the outer surface of the later face **1078** to being placed within the recess **1038**. Positioning the securement member **1030** within the recess **1038** may be beneficial as may reduce the changes that the securement member **1030** will be unintentionally be caught on a projection, which may break the securement member **1030**.

(101) The follow discusses accessories that may be used in connection with the portable light **10**. For sake of brevity, the following accessories will only be discussed in connection with portable light **10**. However, it should be understood that the following accessories may be used in connection with any embodiment of the portable light **10**, including the second embodiment **1001** and the third embodiment **2001**. A first accessory **298** may include a flashlight head **300** that can be connected to the second end **18** of the portable light **10**. Specifically, the user may remove the second end assembly **142** from the portable light **10** and replace this assembly **142** with the flashlight head **300**. This flashlight head **300** would be directly connected to the power source **134** and would draw power therefrom. The flashlight head **300** may include a cone shaped body **302**, a reflector **304** positioned within the cone shaped body **302**, an emitter **306** positioned at the base of the reflector **304**, and a switch **308**. This flashlight head **300** allows a user to focus light in a direction that is perpendicular to the lighting elements **100**. During operation, this flashlight head **300** may disable the lighting elements or may be above to functional along with them. This flashlight head **300** may be desirable because it allows the portable elongated light **10** to selectively operate as a lantern (e.g., illuminating the lighting elements **100**) in a first mode and/or operate like a portable flashlight (e.g., illuminating the flashlight head **300**) in a second mode.

(102) A second accessory may include a speaker that can be connected the second end **18** of the portable light **10**. Specifically, the user may remove the second end assembly **142** from the portable light **10** and replace this assembly **142** with a speaker. This second accessory would be directly connected to the power source **134** and would draw power therefrom. The second accessory may include a wireless module, a one/off button, and a pairing button. The wireless module contained within the speaker allows a user to connect a portable device, like a cellular phone or laptop, in order for the speaker to receive and play audio signals. The on/off button turns on and off the

speaker and the pairing button sends a signal to the wireless module to facilitate the pairing of the wireless module with the portable device. This second accessory may have the same diameter or a larger diameter than the portable light **10**. This second accessory enables the user to listen to music, while utilizing the light that is emitted from the portable light **10**.

(103) A third accessory may include a sensor, a wireless module, or a sensor and wireless module that can be connected the second end **18** of the portable light **10**. Specifically, the user may remove the second end assembly **142** from the portable light **10** and replace this assembly **142** with a sensor or wireless module. This third accessory would be directly connected to the power source **134** and would draw power therefrom. The sensor may be motion, light, pressure, moisture, acceleration, sound, or any combination of these sensors. The wireless module may be compatible with Bluetooth, NFC, Felica, WiFi, Zigbee, RFID, cellular, WiMAX, ISM, or any combination of these technologies. For example, if a local light sensor is contained within the third accessory, the portable light **10** may un-illuminate the lighting elements **100** in response to detecting light that is local to the portable light **10**. In another example, if a local sound sensor (e.g., a microphone) is contained within the third accessory, the user may use voice commands to turn on the portable light **10**. Alternatively, if a wireless module is attached to the portable light **10**, then the elongated light can be controlled by a remote device (e.g., cellular phone, laptop, RF remote control, wireless camera, remote motion sensor, remote light sensor, timer, or remote devices described in U.S. patent application Ser. No. 15/812,852, filed on May 17, 2018, and which is fully incorporated herein by reference). This third accessory enables the control of the portable light **10** by either remotely by a user or by sensors, which may be desirable in a number of situations.

(104) The fourth accessory may include a tripod. Specifically, the user may remove the second end assembly **142** from the portable light **10** and replace this assembly **142** with a connector that can attach to a tripod. This accessory may be beneficial for work at construction sites or a photo shoot.

(105) The fifth accessory may include a charging base. Specifically, the user may remove the second end assembly **142** from the portable light **10** and replace this assembly **142** with second end assembly that includes an usb receptacle or multiple usb receptacles. Specifically, the receptacles may be positioned in the current location of the magnet **190** or they may be positioned on the surface of the bottom housing **186** that is perpendicular to the bottom surface **199**. The positioning on this lateral surface may be beneficial because it would allow a user to place the second end **18** on the support surface, while having access to the usb receptacle. This fifth accessory would be directly connected to the power source **134** and would draw power therefrom.

(106) The sixth accessory may include different types or spare power sources **134**. Accordingly, this sixth accessory may be a removable non-rechargeable battery, a combination of removable non-rechargeable batteries, a removable rechargeable battery, a combination of removable rechargeable batteries, a combination of removable rechargeable batteries disposed within a battery cartridge **250**, a non-removable rechargeable battery, a combination of non-removable rechargeable batteries disposed within a battery cartridge **250**, a DC power supply that is configured to connect to a 12 volt car battery, a DC power supply that is configured to connect to a 110 volt alternative current outlet, or any other type of power supply that is known to a person of skill in the art.

(107) The seventh accessory may include any combination of accessories one through sixth. For example, it may be desirable to have a portable light **10** that has a rechargeable battery that includes an USB **310** receptacle thereon and the bottom housing may also have an USB receptacle **310**. This would allow the user to charge the rechargeable battery using an USB cable. After the battery is charged and placed within the portable light **10**, the user can then charge their cellular phone from the portable light **10** without removing the battery from the light. Alternatively, it may be desirable to use a tripod in connection with a power source that includes an AC-DC converter. Other embodiments or combinations are contemplated by this disclosure.

(108) The portable light **10** enables numerous benefits over prior lighting systems. Unlike a lighting system that merely offers a single mounting configuration, luminosity mode and/or

operational mode, the present disclosure provides for a system that synergistically and advantageously combines at least all of these features to create a unified system providing flexibility, portability and varied operational mode capabilities.

(109) While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that the teachings may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all applications, modifications and variations that fall within the true scope of the present teachings. Other implementations are also contemplated.

(110) While some implementations have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the disclosure; and the scope of protection is only limited by the scope of the accompanying claims. For example, the overall shape of the portable light **10** may be a triangular prism, a rectangular prism, a cube, a pentagonal prism, a hexagonal prism, octagonal prism, sphere, a cone, a tetrahedron, a cuboid, a dodecahedron, a icosahedron, a torus, an octahedron, an ellipsoid, or any other similar shape. Specifically, the portable light **10** can have a triangular or spherical shape and includes an arrangement of a plurality of ribs **150**, lighting elements **100**, as well as the mode selector assembly **32** and the luminosity selector assembly **28**. In this embodiment, the mode selector assembly **32** and the luminosity selector assembly **28** may be depressible buttons. Similarly, the portable light **10** may be configured to have a hexagonal prism shape, where there are six lighting elements **100**. Specifically, the lighting elements **100** may be positioned on the sides of the prism and the ribs **150** may be positioned on the vertices. Further, the mode selector assembly **32** and the luminosity selector assembly **28** may be positioned on the top of the hexagonal prism.

(111) The portable light **10** may have a length that is between 14 and 4 inches, preferably between 12 and 6 inches, and most preferably between 10 and 8 inches. The portable light **10** may have a width that is between 5 and 0.5 inches, preferably between 3 and 1 inches, and most preferably between 2 inches and 1.5 inches. For example, the portable light **10** may have a length that is approximately 8 times the width, preferably 6 times the width, and most preferably at least 5 times the width. In a commercial embodiment, the portable light **10** may have a length that is approximately 9.5 ± 1 inches and a diameter that is approximately 1.7 ± 0.5 inches. This configuration allows the portable light **10** to have an elongated configuration, as the length is substantially greater than the width.

(112) In an embodiment where the portable light **10** has a diameter that is 1.8 inches and has four evenly spaced lighting elements **100**, the arc length of each lighting element **100** may be between 0.4 and 0.2 inches, preferably between 0.35 and 0.25, and most preferably between 0.32 and 0.28. In an embodiment where the portable light **10** has a diameter that is 1.8 inches and the retractable stabilization assembly **192** is in the retracted position **210**, the outer diameter of the retractable stabilization assembly **192** is 1.8 inches. In other words, when the retractable stabilization assembly **192** is in the retracted position **210**, the retractable stabilization assembly **192** has an outer diameter that is the same as the diameter of the portable light **10**. In an embodiment where the portable light **10** has a diameter that is 1.8 inches and the retractable stabilization assembly **192** is in the deployed position or extended position **212**, the outer diameter of the retractable stabilization assembly **192** is between 2.5 inches and 5 inches. In other words, when the retractable stabilization assembly **192** is in the deployed position **212**, the retractable stabilization assembly **192** has an outer diameter that is larger outer diameter than the portable light **10**. In some embodiments, when the retractable stabilization assembly **192** is in the deployed position **212**, the retractable stabilization assembly **192** has an outer diameter that is 4 times larger than the outer diameter than the portable light **10**, preferably 3 times larger than the outer diameter than the portable light **10**, and most preferably at least 1.5 times larger than the outer diameter than the portable light **10**.

(113) In an alternative embodiment, the mode selector assembly **32** and the luminosity selector assembly **28** may be combined into a single assembly that controls both the luminosity and the operational mode of the portable light **10**. For example, this combination assembly may be one or more buttons, switches, sliders, local sensors (e.g., motion, light, sound, heat, motion), or other types of electrical selection devices. In one embodiment, this combination assembly may be a single button that allows the user to select the illumination mode and the operation mode of the portable light **10** by utilizing a combination of depressing and/or holding the button in a depressed state. For example, the user: (i) may depress the button one time to turn “On” on a single lighting element **100**, (ii) may depress the button three time to turn “On” on a three lighting element **100**, and (iii) depress and hold the button a predetermined amount of time to alter the brightness or luminosity level. Additionally, this single assembly may be remotely located from the portable light **10**. Accordingly, in this configuration, the portable light **10** would not have any features on the portable light **10** itself that would allow a user to control the operation of the lighting elements **100**. This configuration may allow the portable light **10** to be waterproofed for up to 30 meters. For example, the luminosity selector assembly **28** may be located on a remote wireless device, such as the remote devices described in U.S. patent application Ser. No. 15/812,852, filed on May 17, 2018, and which is fully incorporated herein by reference. Another example of a remote wireless device includes a cellphone, laptop, RF remote control, or other devices that are connected to the portable light **10** via the internet (e.g., wireless camera, motion sensor, light sensor, timer, etc.).

(114) In one embodiment, the emitter assembly **164** may be a COB LED, which does not need a secondary optic, they have good heat dissipation, and can have a higher density to form a more uniform/higher lumen light. It should be understood that different types of emitter assembly **164** may be utilized, such: (i) a standard LED, (ii) organic LED, (iii) induction light panel, (iv) silicon quantum dot phosphor (SiQD-phosphor), or (v) surface-mount device LED. Further, it should be understood that a combination of the different types of emitter assembly **164** may be utilized. For example, one light emitter assembly **164** may be a surface mounted LED, while other light emitters **164** may be COB LEDs. It should also be understood, that lens **168** may also be omitted. In this configuration, there is no primary optic, no secondary optic, and no reflector.

(115) Depending on type of selected emitter assembly **164** and the light distribution pattern, the shape of the lens **168** may vary. For example, as shown in FIGS. **10-11**, the lens **168** may have a substantially uniform thickness across the emitter assembly **164**. It should be understood that while the lens **168** may continue to have a substantially uniform thickness, the lens **168** may have a greater curvature or less curvature, which in turn will provide a more focused or less focused light distribution pattern, respectively. It should be understood that the outer surface **174** of the lens **168** may have a different configuration, where the lens **168** does not have a substantially uniform thickness. For example, the outer surface **174** of the lens **168** may have a greater curvature than the inner surface **172**. Other thickness variations and/or curvatures combinations are contemplated by this disclosure. For example, some of the lighting elements **100** may have a lens **168** that has a greater curvature, while other lighting elements **100** may have a lens **168** that has a greater curvature. This configuration allows the user to further choose the desired light distribution pattern.

(116) The upper housing **26**, intermediate housing **140**, and the bottom housing **185** may be combined into a single housing and the power source may be a rechargeable battery that is not intended to be removed from the housing. The upper housing **26**, intermediate housing **140**, and bottom housing **186** may be formed from metal, such as aluminum or steel, a polymer material, such as plastic, or a combination of metal and a polymer material. The lens **168** may be made from a polymer material, such as plastic, and may be transparent, partially-transparent, or colored. Some or all of the ribs **150** may be made from a magnet material. The outer rim **40** may also be made from a magnet material.

(117) All elements of the portable light **10** may be colored black except for: i) selection indicator **108**, ii) mode indicia **110**, iii) cap **48**, iv) luminosity indicator **62**, v) lighting elements **100**, vi)

connector ring **36**, and connector ring **182**. In particular, the connector rings **36**, **182** may be colored green. The outer surface **176** of the ribs **150** may include indicia, such as the manufacturer of the light **10** or may be personalized to include a person's name or information.

(118) Headings and subheadings, if any, are used for convenience only and are not limiting. The word exemplary is used to mean serving as an example or illustration. To the extent that the term include, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

(119) Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

(120) Numerous modifications to the present disclosure will be apparent to those skilled in the art in view of the foregoing description. Preferred embodiments of this disclosure are described herein, including the best mode known to the inventors for carrying out the disclosure. It should be understood that the illustrated embodiments are exemplary only, and should not be taken as limiting the scope of the disclosure.

Claims

1. A portable light configured to provide distinct operating modes, the portable light comprising: a central body having a receiver and a central axis that extends through a length of the central body; a power source positioned within the receiver of the central body; a flashlight head operably coupled to the power source and configured to emit light, wherein a portion of the emitted light is oriented substantially parallel with the central axis; an upper housing coupled to the central body and having a mode indicia; a mode selector having: (i) a mode selector ring that circumscribes the upper housing, and (ii) a mode selection indicator; wherein an application of an angularly directed force on the mode selector ring causes said mode selector ring to be angularly displaced relative to the central axis and the upper housing, and wherein said angular displacement of the mode selector ring causes: (i) the mode selection indicator to move relative to the mode indicia, and (ii) the portable light to change from a first operational mode to a second operational mode.
2. The portable light of claim 1, wherein the power source is a rechargeable battery configured to: (i) supply current to a USB receptacle, and (ii) receive current from the USB receptacle.
3. The portable light of claim 1, further comprising a button configured to switch the portable light from an OFF state to an ON state, and wherein in the OFF state, no light is emitted from the flashlight head, and wherein in the ON state, light is emitted from the flashlight head.
4. The portable light of claim 1, wherein the mode selector ring is configured to be displaced less than 360 degrees.
5. The portable light of claim 4, wherein the mode selector ring is configured to be displaced less than 100 degrees.
6. The portable light of claim 1, wherein in the first operational mode, the portable light emits light

at a first lumen output level, and wherein in the second operational mode, the portable light emits light at a second lumen output level, the first lumen output level being less than the second lumen output level.

7. The portable light of claim 1, wherein the portable light is waterproof for up to 30 meters.

8. The portable light of claim 1, wherein the portable light is configured to output more than 3,000 lumens.

9. The portable light of claim 1, wherein the mode selector ring includes an external surface that is configured to enhance the user's grip during the angular displacement of the mode selector ring.

10. A portable light configured to provide multiple operating modes, the portable light comprising: a mode selector having a mode selector ring that circumscribes a portion of a housing of the portable light; a light emitter: (a) operatively coupled to the mode selector, (b) configured to emit a portion of light in a direction that is substantially parallel to a central axis of the portable light, and (c) configured to emit (i) a first amount of light when the portable light is in a first operational mode and (ii) a second amount of light when the portable light is in a second operational mode; and wherein the mode selector ring is configured to be rotated relative to the central axis and the housing of the portable light, and wherein said rotation of the mode selector ring causes the portable light to change from the first operational mode to the second operational mode.

11. The portable light of claim 10, further comprising a power source that is coupled to the mode selector and the light emitter, and wherein said power source includes a rechargeable battery configured to: (i) supply current to a USB receptacle, and (ii) receive current from the USB receptacle.

12. The portable light of claim 10, further comprising a button configured to switch the portable light from an OFF state to an ON state, and wherein in the OFF state, no light is emitted from the flashlight head, and wherein in the ON state, light is emitted from the flashlight head.

13. The portable light of claim 10, wherein the mode selector ring is configured to be rotated less than 100 degrees.

14. The portable light of claim 10, wherein in the first operational mode, the portable light emits light at a first lumen output level, and wherein in the second operational mode, the portable light emits light at a second lumen output level, the first lumen output level being less than the second lumen output level.

15. The portable light of claim 10, wherein the mode selector ring includes a means for enhancing the user's grip during rotation of the mode selector ring.

16. A portable light comprising: a housing coupled to a central body, and wherein said central body includes a receiver and a central axis that extends along a length of the central body; a power source positioned within the receiver of the central body; a flashlight head coupled to an extent of the power source and configured to emit: (i) a first amount of light in a first mode and (ii) a second amount of light in a second mode, and wherein the second amount of light is greater than the first amount of light; a mode selector ring that circumscribes a portion of the central body; and wherein the mode selector ring is configured to be rotated around the central axis and the central body, and wherein said rotation of the mode selector ring causes the portable light to change between the first mode and the second mode.

17. The portable light of claim 16, wherein the power source is a rechargeable battery configured to: (i) supply current to a USB receptacle, and (ii) receive current from the USB receptacle.

18. The portable light of claim 16, further comprising a button configured to switch the portable light from an OFF state to an ON state, and wherein in the OFF state, no light is emitted from the flashlight head, and wherein in the ON state, light is emitted from the flashlight head.

19. The portable light of claim 16, wherein the mode selector ring is configured to be rotated less than 100 degrees.

20. The portable light of claim 16, wherein the portable light is configured to output more than 3,000 lumens.

